



Society of Petroleum Engineers

SPE-226847-MS

Stratigraphic and Sedimentological Predictions, From Regional to Field Scale Approach: Examples from Aptian Intrashelf Strata, Saudi Arabia

Marco I. Alvarez Bastos, Awadelgied Adam, and Ahmed Awami, Saudi Aramco

Copyright 2025, Society of Petroleum Engineers DOI [10.2118/226847-MS](https://doi.org/10.2118/226847-MS)

This paper was prepared for presentation at the Middle East Oil, Gas and Geosciences Show (MEOS GEO), Manama, Bahrain, 16 – 18 September 2025.

This paper was selected for presentation by an SPE program committee following review of information contained in an abstract submitted by the author(s). Contents of the paper have not been reviewed by the Society of Petroleum Engineers and are subject to correction by the author(s). The material does not necessarily reflect any position of the Society of Petroleum Engineers, its officers, or members. Electronic reproduction, distribution, or storage of any part of this paper without the written consent of the Society of Petroleum Engineers is prohibited. Permission to reproduce in print is restricted to an abstract of not more than 300 words; illustrations may not be copied. The abstract must contain conspicuous acknowledgment of SPE copyright.

Summary

The carbonate ramp model has been commonly used to interpret the depositional settings of the Lower Cretaceous calcareous strata across the Arabian Plate. This model generally fits at the field scale; however, it fails when regional data is integrated across a broader area. As a consequence, this may mislead stratigraphic predictions. The integration of physical stratigraphy, core-based sedimentology, petrophysics, and seismic interpretation helped to build a robust stratigraphic framework and to construct gross depositional environment (GDE) indicators from each subject matter (e.g., 3D strata distribution, gross and net thickness maps, rock types trends, common vertical facies successions and lateral facies associations, and geomorphological seismic character from various attributes). The integrated analysis of GDE indicators helps not only to build comprehensive GDE maps but also to predict additional geomorphologic or paleoenvironmental details. The study herein shows an integrated GDE mapping depicting platform interior geomorphology within associated intrashelf settings for the Aptian strata. Extraction and display of a number of maps, time slices, and cross sections helped to capture observations regarding the 4D distribution of strata showing clastic wedges and shoal/bank complexes across the studied area. The Aptian strata across the AOI display lateral changes from clastic-dominated systems (sandy to muddy) passing into calcareous platform interior systems (muddy to grainy). The calcareous systems are associated with a highly heterogeneous rudist shoal/bank complex. This complex was vertically subdivided in two intermediate frequency base-level cycles. The basal cycle shows a landward stacking pattern, and the overlying cycle shows a seaward (basinward) stacking pattern. At intermediate or local mapping scale, the discrete rudistic shoals may laterally display clinoform-like morphologies at a scale of 10s of km. When shoals are observed isolated and out of regional context, their morphologies may be misinterpreted as a carbonate ramp. This study showcases 2D and 3D diagrams to illustrate the architecture and distribution of the intrashelf Aptian rudist shoal/bank complexes and their implication for stratigraphic and sedimentologic predictions. The findings of this study helped to recognize the widespread and overuse of ramp models throughout many stages, including the Aptian, across the Arabian Peninsula. In the past, this localized view and interpretation of the data have prevented geoscientists from making reasonable stratigraphic predictions. Herein, we appeal for an integrated GDE mapping approach to further enhance predictions and unveil stratigraphic exploration concepts across the Arabian Plate.

Keywords: sequence stratigraphy, stratigraphic predictions, regional, Aptian, intrashelf basins, Arabia

Introduction

The carbonate ramp model, first introduced by [Ahr \(1973\)](#), is a depositional model that describes a gently sloping surface with a gradual transition from shallow, wave-agitated waters to deeper, lower-energy environments. According to [Burchette and Wright \(1992\)](#), the term "ramp" refers to a depositional surface with a slope of less than 1° , characterized by a continuous and gradual offshore transition without a distinct slope break. This model differs from rimmed shelves, where shoal deposits are often far from the shoreline. In contrast, the carbonate ramp model predicts that shoal deposits will occur near the shoreline. [Burchette and Wright \(1992\)](#) noted that the Arabian Gulf, as described by [Purser \(1973\)](#), appears to represent a modern environment that conforms to the ramp ideal. As a result, the carbonate ramp model has been widely adopted as a reference for the Arabian Peninsula and has been extensively used to interpret the depositional settings of the Lower Cretaceous calcareous strata across the Arabian Plate. The model's simplicity and apparent relevance to the regional geology have made it a popular choice for describing the environmental setting in different areas. Numerous authors, including [Burchette and Wright \(1992\)](#) and more recently, [Rivers and Dalrymple \(2025\)](#), have discussed, refined, and enhanced the ramp model.

A comprehensive review of the stratigraphic literature on the Arabian Plate reveals the widespread adoption of the ramp model to describe the occurrence of carbonates across various stages. This model has been extensively applied in numerous studies, including those by [Amthor et al. \(2010\)](#), who utilized it for reservoir characterization of a Shu'aiba carbonate ramp-margin field in northern Oman ([Figure 1](#)). Similarly, [Droste et al. \(2010\)](#) employed the ramp model to discuss the sequence-stratigraphic framework of the Aptian in Oman, while [Droste \(1990\)](#) applied it to examine depositional cycles and source rock development in the Arabian Peninsula. [Lindsay et al. \(2006\)](#) also used the model to describe shoaling-upward carbonate cycles in Arabia, and more recently [Schlaich and Aigner \(2017\)](#) while describing the Dhurma Formation in Oman, and [Al-Ghamdi and Read \(2010\)](#) invoked ramp terminology in their subregional discussion of the Aptian ([Figure 2](#)), describing the facies-based sequence-stratigraphic framework of the Lower Cretaceous rudist platform in Saudi Arabia. The common application of the ramp model in these studies highlights its overuse in understanding the stratigraphic evolution of the Arabian Plate.

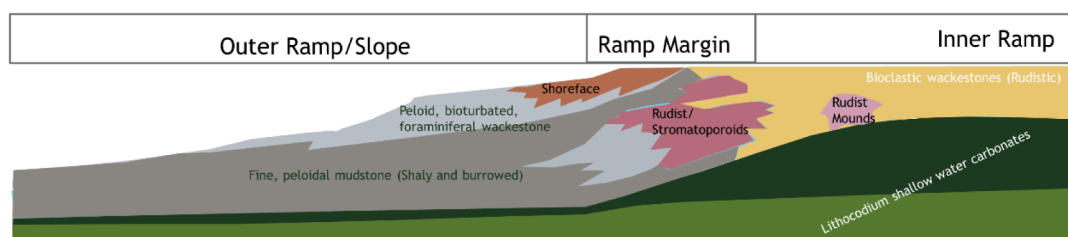


Figure 1—Conceptual ramp-to-basin profile for Aptian carbonate strata in Oman. This schematic illustration, adapted from [Amthor et al. \(2010\)](#), depicts a typical ramp-to-basin profile, showcasing the stratigraphic evolution of carbonate strata during the Aptian stage in Oman. The profile highlights the characteristic depositional environments and facies relationships that are commonly interpreted in this region.

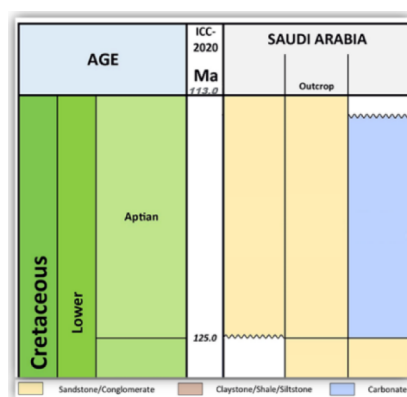


Figure 2—Schematic lithologic nature of the stratigraphic record for the Aptian strata across Saudi Arabia. Observe the conspicuous distribution of clastics and carbonates over a 12-million-year period.

Alternative perspectives for the Aptian strata regarding the depositional settings of the Arabian Plate have been proposed by several researchers, offering a more complex and nuanced understanding of the region's geological history. For instance, [McGuire et al. \(1993\)](#) characterized the Middle Jurassic Arabian Platform as a region with minimal bathymetric relief and widespread shallow-marine carbonate sedimentation, where intrashelf basins, and intervening arches developed. Similarly, [Murriss \(1980\)](#) discussed the stratigraphic evolution and oil habitat in the Middle East, describing the formation, evolution, and scale of intrashelf basins, and referencing the "rimmed shelf model" of [Wilson \(1975\)](#) in relation to the Aptian depositional settings. Furthermore, [Ziegler \(2001\)](#) provided valuable insights into the regional distribution and nature of intrashelf basins, and the potential impact of structural factors on sedimentation.

Building on these alternative perspectives, this study adopts a platform-interior and intrashelf-setting framework to describe the observations for the Aptian strata collected from multiple areas of study. By doing so, we aim to provide a more comprehensive and detailed understanding of the depositional settings of the Arabian Plate, moving beyond the limitations of the traditional carbonate ramp model and offering a more integrated representation of the region's geological complexity.

Method

This study presents a comprehensive analysis of the Aptian strata in the study areas, highlighting the key observations and insights gained from the investigation. The lithologic nature of the stratigraphic record for the Aptian strata is characterized by a distinctive distribution of carbonates and a reciprocal relationship with clastic sediments over a period of 12 million years ([Figure 2](#)), which is influenced by structural constraints. To develop a robust stratigraphic framework, we integrated multiple disciplines, including physical stratigraphy, core-based sedimentology, petrophysics, and seismic interpretation. This multi-faceted approach enabled us to construct gross depositional environment (GDE) indicators from various data sources, such as 3D strata distribution, gross and net thickness maps, rock type trends, common vertical facies successions, lateral facies associations, and geomorphological seismic character derived from various attributes.

The integrated analysis of GDE indicators not only allowed us to create comprehensive GDE maps but also facilitated the prediction of additional geomorphological and paleoenvironmental details. [Figure 3](#) presents a schematic workflow illustrating the key steps involved in this process. This workflow highlights the importance of integrating multiple data sources and analytical techniques to shed light on the stratigraphic framework of the studied strata and to gain a deeper understanding of the depositional environments, geomorphology and the stratigraphic framework of the Aptian strata.

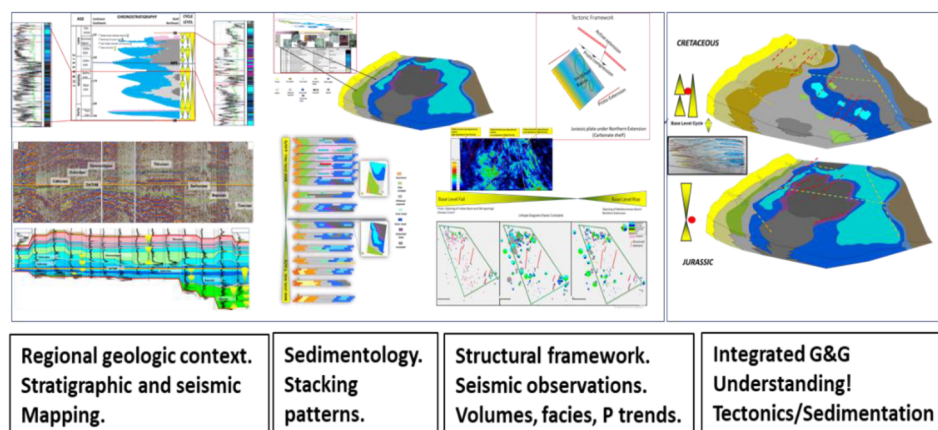


Figure 3—Schematic workflow illustrating the key steps involved in elucidating the stratigraphic framework and associated structural constraints of the studied strata (Modified from Alvarez, 2024).

Results

This study utilizes 2D and 3D diagrams to illustrate the complex architecture and distribution of intrashelf Aptian rudist shoal/bank complexes, providing valuable insights for stratigraphic and sedimentologic predictions. An integrated Gross Depositional Environment (GDE) mapping approach is employed to depict the platform interior geomorphology within associated intrashelf settings for the Aptian strata.

Figure 4 presents a representative example of typical facies and facies associations characterized from cores in Aptian strata. The characteristic cycles exhibit a base-level rise hemicycle, marked by a basal ravinement deposit and capped by muddy facies. In contrast, the base-level fall is characterized by a transition from mudstones to wackestones, and from muddy packstones to packstones and grainstones, including floatstones and rudstones. By analyzing the proportion of facies types, a combination of grainstones and packstones or rudstones and floatstones indicates inner shoal or bank settings. On the other hand, an increasing proportion of wackestones, associated with floatstones or rudstones, is assigned to the backshoal or foreshoal facies tract. The conspicuous association of mudstones and muddy facies is characteristic of lagoonal settings. Further inland, the rhythmic interbedding of wavy laminated sandy and shaly facies is associated with deltaic settings.

The geomorphologic concept of the shoals/banks is further illustrated by the patchy, undulating, and wavy nature of the Belize model analog, extracted from Google Maps. This analog provides a valuable visual representation of the complex architecture and distribution of the intrashelf Aptian rudist shoal/bank complexes, highlighting the importance of integrated GDE mapping and facies analysis in understanding the stratigraphic and sedimentologic evolution of these systems.

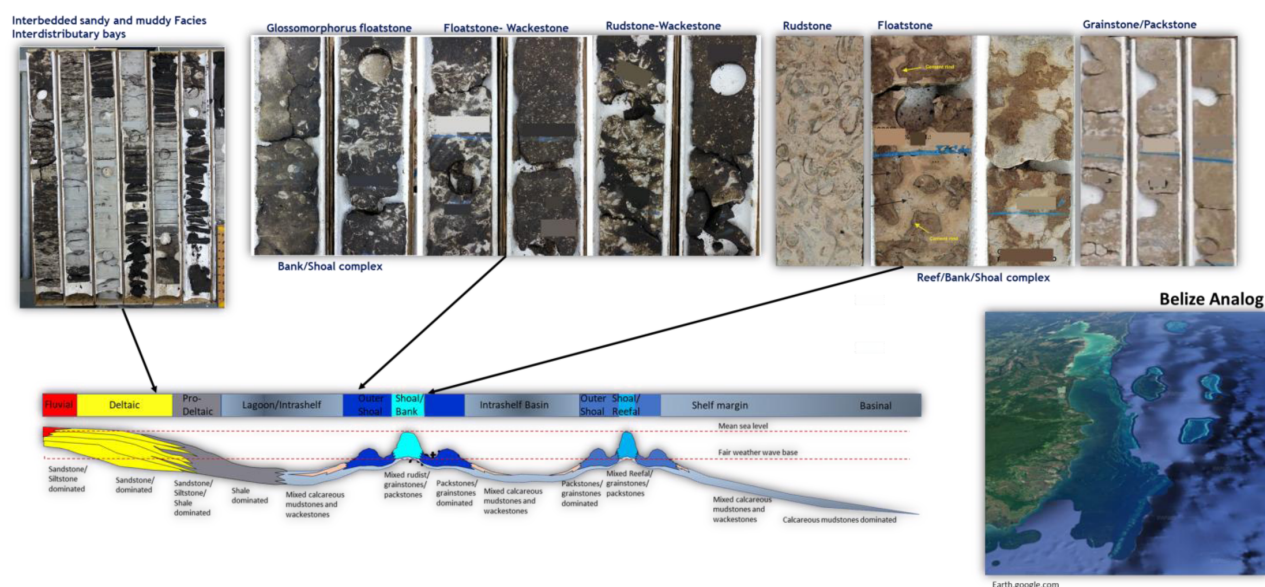


Figure 4—Typical facies and facies associations characterized from cores in Aptian strata. Note the facies diversity of strata. In 2D, application of a simple ramp model constrained by 4D facies recognition and spatial mapping and predictability of facies tracts would fail. Observe the patchy and undulating/wavy nature of the shoals concept capture from the Belize model analog (Google maps).

Figure 5 presents a schematic 2D depositional model illustrating the main environmental settings within a shelf, from fluvial to deltaic conditions, deepening to "offshore" and restricted lagoonal settings. The carbonate factory is represented by a restricted lagoon that changes into a backshoal and shallows up to inner shoals, and then deepens to foreshoals and ultimately to deep lagoon or intrashelf settings. The system then shallows up from deep lagoons or intrashelf conditions to backshoal, and shallows further into shoals/reefal settings, where the shelf margin and slope changes may be recognized. These settings deepen to foreshoal/forereef and rapidly change into basin margin and open, deeper marine basinal settings. Notably, peritidal settings can be associated with the clastic wedges and the carbonate factory, particularly around the foreshore, foreshoal, or backshoal settings.

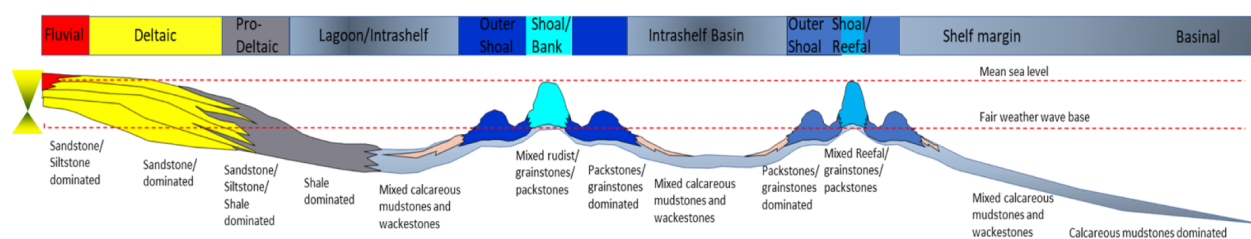


Figure 5—Schematic 2D depositional model showing main environmental settings within a shelf. Note that intrinsic to this diagram, the peritidal settings can be associated with the clastic wedges and the carbonate factory. Additionally, observe the spatial geometry and areal relationships among sub-environments. Note that the application of localized models or out of a regional geologic context (i.e., intrashelf settings) would lead to indiscriminate use of ramp models fitting localized observations but failing to explain regional areal distribution of strata, volume partitioning and facies differentiation.

However, it is essential to consider the spatial geometry and areal relationships among sub-environments in a regional geologic context, as localized studies may lead to incorrect estimates of changes in slope and proposed open sea directions. The application of localized models without consideration of the regional geologic context can result in the indiscriminate use of ramp models that fit localized observations but fail to explain regional depositional settings, such as platform interior and associated intrashelf settings, areal distribution of strata, volume partitioning, facies differentiation, and potential structural constraints on sedimentation.

The Aptian strata across the study area exhibit lateral changes from clastic-dominated systems (sandy to muddy) to calcareous platform interior systems (muddy to grainy), which are associated with a highly heterogeneous rudist shoal/bank complex. The Aptian carbonate formation can be vertically subdivided into two intermediate-frequency base-level cycles. The basal cycle displays a landward stacking stratigraphic pattern (LST), characterized by a clear shift and expansion of the highly compartmentalized carbonate factory across underlying clastic-dominated settings and patchy distributed carbonates. In contrast, the overlying cycle exhibits a seaward (basinward) stacking stratigraphic pattern (SST), documented by the redistribution and amalgamation of the rudistic banks and associated back-banks and restricted lagoon settings.

At an intermediate or local mapping scale, the discrete rudistic shoals and mounds may laterally display prograding clinoform-like morphologies at a scale of tens of kilometers. However, when these geometries are observed in isolation and out of regional context, at scales of tens to hundreds of kilometers, their regional morphologies may be misinterpreted as a carbonate ramp. Consequently, applying a simple ramp model, constrained by 4D facies recognition, spatial mapping, and predictability of facies tracts across a broad scale, would be inadequate and may lead to incorrect interpretations.

The extraction and display of multiple seismic traverses and stratigraphic cross-sections, including isopach, net reservoir maps, time slices, and seismic attribute map displays (Figure 6), enabled the capture of key observations and the generation of an integrated Gross Depositional Environment (GDE) map (Figure 7). By analyzing the seismic attributes (RMS amplitude) data in cross-section, a subregional-scale glimpse of the seismic character of the Aptian strata was obtained, revealing a high degree of heterogeneity and apparent unidirectional changes in depositional settings. However, the map view display provided a broader perspective on the heterogeneity, multiscale self-similarity, and fractal nature of the carbonate system, which helped to falsify the localized use of the ramp model in the study area.

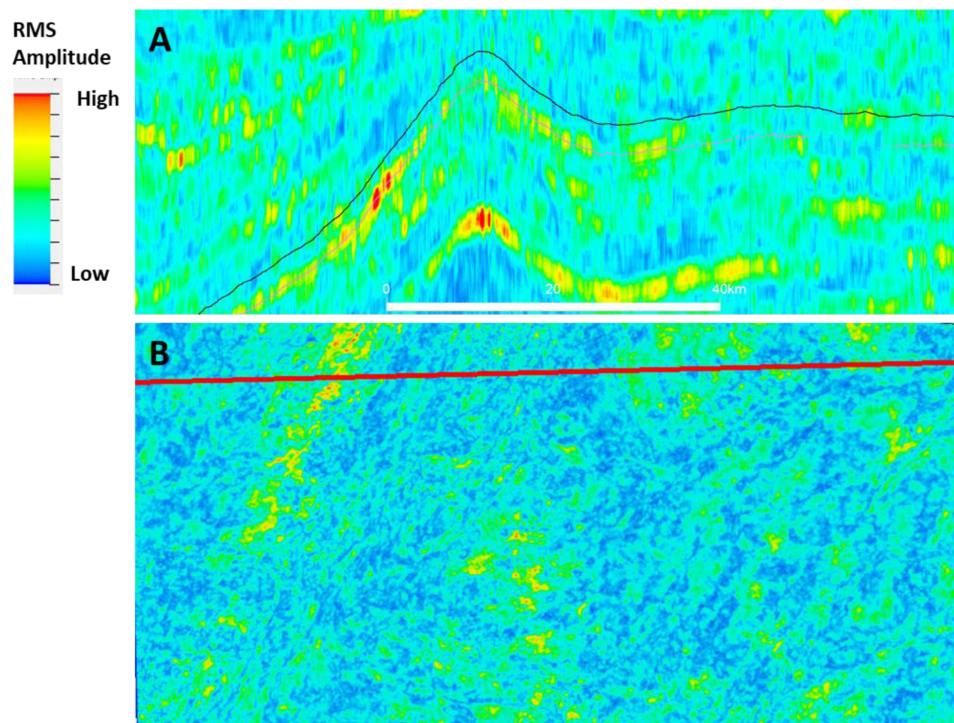


Figure 6—Seismic attributes, specifically RMS amplitude, reveal the seismic character of the Aptian strata at a subregional scale, exposing a highly heterogeneous system. Notably, the data exhibit a lack of unidirectional changes in depositional settings (A), which contradicts the assumptions of the ramp model. This map view of the same data (B) helps to clarify the observations and falsify the localized application of the ramp model in the study area, highlighting the need for a more nuanced and integrated understanding of the depositional environment.

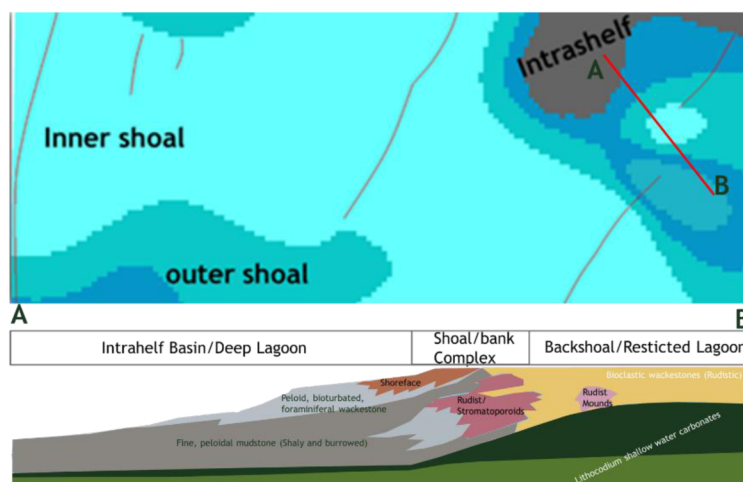


Figure 7—Gross Depositional Environment (GDE) map for the Aptian strata in a localized area, providing a regional context for the platform interior and associated intrashelf basins. Notably, the map reveals the possibility of identifying a line of section that, when viewed in isolation, could be misinterpreted as a ramp model (as shown in [Figure 1](#)). However, when considered within the broader regional context, it becomes clear that the actual depositional environment is more complex and heterogeneous, comprising a platform interior with associated intrashelf basins.

The conceptual GDE map, shown in [Figure 7](#), further illustrates the areal changes in environmental settings across the Aptian strata in one of the study areas. Notably, the rigorous integration of seismic attributes with the GDE suggested additional heterogeneity that needs to be addressed and considered in detailed mapping. The integration of multiple workflows and observations facilitated the conceptualization of the integrated 4D distribution of strata, including clastic wedges and shoal/bank complexes, across the studied areas.

[Figure 8](#) presents the integrated 4D depositional model for the studied Aptian strata, which evolved from the 2D depositional model and integrated workflows and analyses. The generation of such models enables the delivery of a comprehensive model that simplifies multi-directional and localized models, providing a framework for stratigraphic predictions and exploration concepts. The integrated model facilitates the delivery of the areal distribution of strata and unveils potential exploration concepts, ultimately enhancing our understanding of the Aptian strata and informing future exploration strategies.

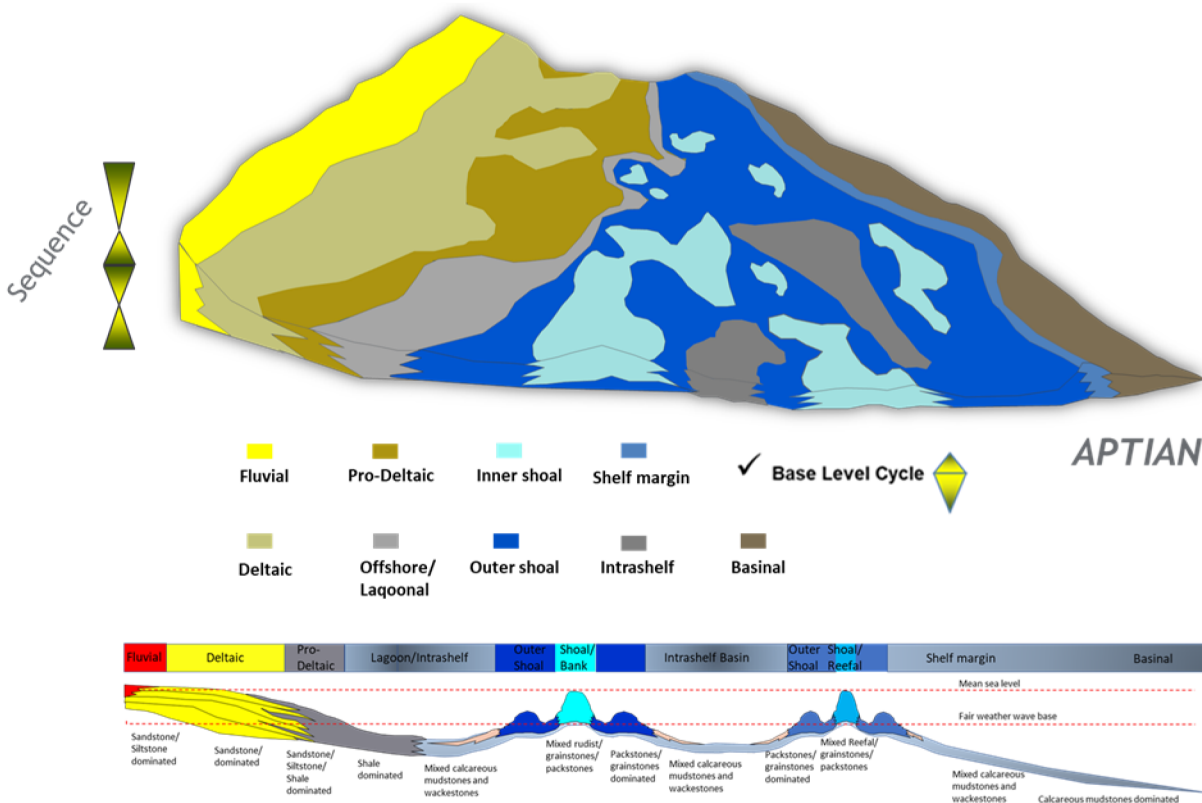


Figure 8—The integrated depositional model for the studied Aptian strata represents a significant advancement in our understanding of the depositional environment, as it evolves from a 2D conceptual model to a comprehensive 4D integrated model. This integrated model seamlessly combines multiple datasets and observations, facilitating the delivery of a robust and accurate representation of the areal distribution of strata. By simplifying multi-directional and localized models into a single, cohesive framework, the integrated model provides a powerful tool for stratigraphic predictions and the identification of potential exploration concepts.

Discussion and Conclusions

The carbonate ramp model has been widely employed to interpret the depositional settings of the Lower Cretaceous calcareous strata across the Arabian Plate, including many other multiple stages. However, this study adopts a more nuanced approach, utilizing the concept of platform interiors and intrashelf settings to describe and interpret the integrated observations collected across multiple carbonate systems, including the Aptian strata in Saudi Arabia. By doing so, we aim to move beyond the limitations of the traditional carbonate ramp model and provide a more comprehensive understanding of the depositional settings of the Arabian Plate.

Our research highlights the importance of understanding the structural evolution of the Arabian Plate and the sub-regional controls on sedimentation through time, which is crucial for envisioning the stratigraphic heterogeneity captured in the stratigraphic record. These structural controls on sedimentation reveal the interplay between clastic wedges and the carbonate factory and help to understand the regional and sub-regional variation of sub-environments, their self-similar nature, and stratigraphic evolution in time and space.

The findings of this study reveal the widespread and overuse of ramp models throughout many stages, including the Aptian, across the Arabian Peninsula. In the past, this localized view and interpretation of the data have prevented geoscientists from making accurate stratigraphic predictions. Therefore, we advocate for an integrated Gross Depositional Environment (GDE) mapping approach, which depicts platform interiors and associated intrashelf basins, to enhance predictions and unveil stratigraphic exploration concepts across the Arabian Plate. This approach will enable geoscientists to better understand the

complexity of these depositional settings and make more informed decisions regarding exploration concepts and strategies.

Acknowledgments

We would like to extend our appreciation to our Aramco colleagues, whose valuable insights, discussions, and contributions have been instrumental in shaping the concepts and ideas presented in this paper. Their motivation, expertise, and collaborative spirit have been essential in facilitating the successful completion of this research, and we are grateful for their involvement and support.

References

- Ahr, W.M., 1973. The Carbonate Ramp-An Alternative to The Shelf Model. *Trans., Gulf Coast Assoc. Geot. Soc.*, **23**: 221–225.
- Al-Ghamdi N., Read J., 2010. Facies-Based Sequence-Stratigraphic Framework of The Lower Cretaceous Rudist Platform, Shu'aiba Formation, Saudi Arabia. In *Barremian-Aptian Stratigraphy and Hydrocarbon Habitat of The Eastern Arabian Plate*. GeoArabia Special Publication, **4**, pp.367–410.
- Amthor, J., Kerans, C., and Gauthier, P., 2010. Reservoir characterization of a Shu'aiba Carbonate Ramp Margin Field, North Oman. GeoArabia Special Publication. **2**. 549–576.
- Alvarez-Bastos, M., 2024. Tectono-stratigraphic constraints in Carbonate and Mixed Carbonate Intrashelf Basins, Saudi Arabia. In *AAPG's Cross Regional Carbonates and Mixed Carbonate Systems Symposium*. Palermo, Italy. Memoirs, p10.
- Burchette, T.P. and Wright, V.P., 1992. Carbonate Ramp Depositional Systems. In B.W. Seliwood (Editor), *Ramps and Reefs*. Sediment. Geol., **79**: 3–57.
- Droste, H.J., Van Buchem, F.S.P., Al-Husseini, M.I. and Maurer, F., 2010. Sequence-Stratigraphic Framework of The Aptian Shu'aiba Formation in The Sultanate of Oman. In *Barremian-Aptian stratigraphy and hydrocarbon habitat of the eastern Arabian Plate*. GeoArabia Special Publication, **4**, pp.229–283.
- Droste, H. (1990) Depositional cycles and source rock development in an epeiric intra-platform basin: the Hanifa Formation of the Arabian Peninsula. *Sed. Geol.*, **69**, 281–296.
- McGuire, M.D., Koepnick, R.B., Markello, M.L., Stockton, M.L., Waite, L.E., Kompanik, M.J., Al-Shammary, M.J., AlAmoudi, M.O., 1993. Importance of Sequence Stratigraphic Concepts in Development of Reservoir Architecture in Upper Jurassic Grainstones, Hadriya and Hanifa Reservoirs, Saudi Arabia. *Society of Petroleum Engineers*, 489–499.
- Lindsay, R., Cantrell, D., Hughes, G., Mueller, T., Russell, D., 2006. Ghawar Arab-D Reservoir: Widespread porosity in shoaling-upward carbonate cycles, Saudi Arabia. In P. M. Harris and L. J. Weber, eds., *Giant hydrocarbon reservoirs of the world: From rocks to reservoir characterization and modeling: AAPG Memoir 88/SEPM Special Publication*, p. 97–137.
- Murris, R.J. 1980. Middle East Stratigraphic Evolution and Oil Habitat. *American Association of Petroleum Geologists Bulletin*, v. **64**, p. 597–618.
- Purser, B., 1973. Sedimentation around bathymetric highs in the southern Persian Gulf. In: *The Persian Gulf: Holocene Carbonate Sedimentation and Diagenesis in a Shallow Epicontinental Sea*. Springer-Verlag, Berlin, pp. 157–177.
- Rivers, J.M., and Dalrymple, R.W., 2025, The reconstruction of coastal carbonate sequence stratigraphy: A modern-systems approach: *Geology*, v. **53**, p. 232–237, <https://doi.org/10.1130/G52776.1>
- Schlaich, M., and Aigner, T., 2017. Facies and integrated sequence stratigraphy of an Epeiric Carbonate Ramp Succession: Dhurma Formation, Sultanate of Oman. In *The Depositional Record*; **3**(1):92–132.
- Ziegler M.A., 2001. Late Permian to Holocene Paleofacies Evolution of The Arabian Plate and Its Hydrocarbon Occurrences. *GeoArabia* **6** (3), 445–504.
- Wilson, J.L., 1975. *Carbonate facies in geologic history*. Springer (New York); 469pp.